

508 then the first molecule of B binds to EA_M to form the complex EAB_M is formed. This binding is
 509 followed by the binding of substrate C and the formation of the complex $EABC_M$ complex. In the
 510 last binding step, the second molecule of B is bound, and the full complex ($EABCB_M$) is formed
 511 on the M face of the membrane. In this work, the model of the kinetic-transport mechanism is
 512 driven by applying Stein's (2012) parameterization method, and the overall transport fluxes of A,
 513 B, and C are expressed in equations 59b, 59c, and 59d, respectively.

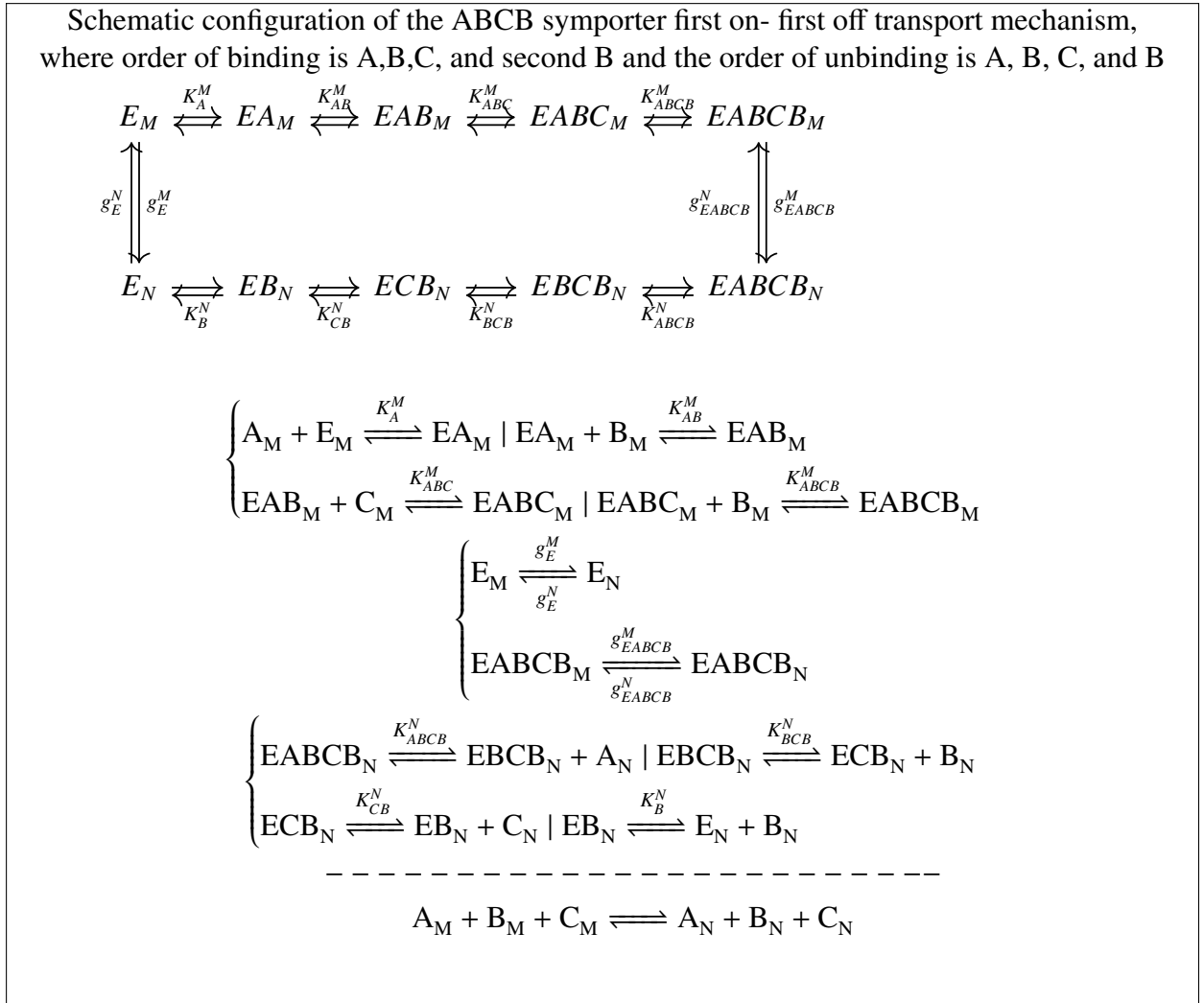


Figure 15: A schematic configuration of the symporter mechanism transporting three solutes (ABCB), and the corresponding net transported flux equation. In this configuration, the carrier exists in 5 different states on each side of the membrane. Dissociation constants (K_A , K_{AB} , K_{ABC} , K_{ABCB} , K_{BCB} , K_{CB} , and K_B terms), and the translocation rate constants (g_E and g_{EABCB} terms) are depicted in the reaction cycle in the right panel. $[E]_t$ is the total number of transporter per unit area of the membrane (part 1/2 continued on next page).

Net Transport Flux from M to N

$$J_{symporter}^{M,N(net)} = [E]_t \left(\frac{(g_{EABC}^M \alpha^M \beta^M \gamma^M \beta''^M) g_E^N - (g_{EABC}^N \alpha^N \beta^N \gamma^N \beta''^N) g_E^M}{R_M R_{NN} + R_N R_{MM}} \right) \quad (59a)$$

$$J_{A,symporter}^{M,N(net)} = J_{symporter}^{M,N(net)} \quad (59b)$$

$$J_{B,symporter}^{M,N(net)} = 2 J_{symporter}^{M,N(net)} \quad (59c)$$

$$J_{C,symporter}^{M,N(net)} = J_{symporter}^{M,N(net)} \quad (59d)$$

where

$$[E]_t = [E]_M + [EA]_M + [EAB]_M + [EABC]_M + [EABC]_N + [EABC]_N + [EAB]_N + [EA]_N + [E]_N$$

$$\alpha^M = \frac{[A]_M}{K_A^M}, \quad \beta^M = \frac{[B]_M}{K_B^M},$$

$$\alpha^N = \frac{[A]_N}{K_A^N}, \quad \beta^N = \frac{[B]_N}{K_B^N}$$

$$\gamma^M = \frac{[C]_M}{K_C^M}, \quad \beta''^M = \frac{[B]_M}{K_{BC}^M},$$

$$\gamma^N = \frac{[C]_N}{K_C^N}, \quad \beta''^N = \frac{[B]_N}{K_{BC}^N}$$

$$R_M = 1 + \alpha^M + \alpha^M \beta^M + \alpha^M \beta^M \gamma^M + \alpha^M \beta^M \gamma^M \beta''^M$$

$$R_N = 1 + \beta''^N + \gamma^N \beta''^N + \beta^N \gamma^N \beta''^N + \alpha^N \beta^N \gamma^N \beta''^N$$

$$R_{MM} = g_E^M + g_{EABC}^M \alpha^M \beta^M \gamma^M \beta''^M$$

$$R_{NN} = g_E^N + g_{EABC}^N \alpha^N \beta^N \gamma^N \beta''^N$$

Figure 15: A schematic configuration of the symporter mechanism transporting three solutes (ABCB), and the corresponding net transported flux equation. In this configuration, the carrier (cotransporter) exists in 5 different states on each side of the membrane. Dissociation constants (K_A , K_{AB} , K_{ABC} , K_{ABCB} , K_{BCB} , K_{CB} , and K_B terms), and the translocation rate constants (g_E and g_{EABC} terms) are depicted in the reaction cycle in the right panel. $[E]_t$ is the total number of transporter per unit area of the membrane (part 2/ 2 continued from last page).

Competitive symporter ordered binding model:

Figure 16 shows a competitive symport transport mechanism involving substrates A , B , and C , in which C competes with B for the same binding sites on the carrier. In the first step, a molecule of A binds to the empty carrier (E), followed by the binding of a B or C molecule to form the complex EAB or EAC , respectively. Next, the formed complexes cross the membrane (EAC_N or EAB_N), where they undergo dissociation reactions in which A unbinds last from both complexes. In this model, it is assumed that the carrier can exist in four different states at each side of the membrane, that only the fully loaded (EAB and EAC) and empty carriers (E) can cross the membrane. The net